

The canonical form only includes “lesser or equal” inequalities. Therefore, we can transform each equality constraint into two inequality constraints.

The first constraint is

$$x_1 + x_2 + x_3 = 1. \quad (1)$$

It can be written with two inequality constraints:

$$\begin{aligned} x_1 + x_2 + x_3 &\leq 1, \\ x_1 + x_2 + x_3 &\geq 1, \end{aligned}$$

or, equivalently,

$$\begin{aligned} x_1 + x_2 + x_3 &\leq 1, \\ -x_1 - x_2 - x_3 &\leq -1. \end{aligned}$$

The second constraint is

$$x_1 - x_2 + x_4 = 1. \quad (2)$$

It can be written with two inequality constraints:

$$\begin{aligned} x_1 - x_2 + x_4 &\leq 1, \\ x_1 - x_2 + x_4 &\geq 1, \end{aligned}$$

or, equivalently,

$$\begin{aligned} x_1 - x_2 + x_4 &\leq 1, \\ -x_1 + x_2 - x_4 &\leq -1. \end{aligned}$$

Finally, the fact that x_i is non negative can be written as $-x_i \leq 0$, for $i = 1, \dots, 4$.

Putting everything together, we obtain

$$\begin{aligned} x_1 + x_2 + x_3 &\leq 1, \\ -x_1 - x_2 - x_3 &\leq -1, \\ x_1 - x_2 + x_4 &\leq 1, \\ -x_1 + x_2 - x_4 &\leq -1, \\ -x_1 &\leq 0, \\ -x_2 &\leq 0, \\ -x_3 &\leq 0, \\ -x_4 &\leq 0. \end{aligned}$$

In matrix form, we have

$$\mathcal{P} = \{x \in \mathbb{R}^4 \mid Ax \leq b\},$$

where

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & -1 & -1 & 0 \\ 1 & -1 & 0 & 1 \\ -1 & 1 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Note that there are other ways to write the polyhedron in canonical form. In particular, if we observe that x_3 and x_4 in the standard form are slack variables, we can eliminate them in the canonical form. Indeed, we can combine (1) with the non negativity constraint of x_3 to obtain:

$$x_3 = 1 - x_1 - x_2 \geq 0,$$

or, equivalently

$$x_1 + x_2 \leq 1.$$

And x_3 disappears from the formulation. Similarly, (2) can be written

$$x_4 = 1 - x_1 + x_2 \geq 0,$$

or, equivalently

$$x_1 - x_2 \leq 1.$$

Putting everything together, we obtain:

$$\begin{aligned} x_1 + x_2 &\leq 1, \\ x_1 - x_2 &\leq 1, \\ -x_1 &\leq 0, \\ -x_2 &\leq 0. \end{aligned}$$

In matrix form, it is:

$$\mathcal{P} = \{x \in \mathbb{R}^2 \mid Ax \leq b\}$$

with

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

This is now a polyhedron in $\mathbb{R}^2$.